

Ask the word. Never the *sentence*.

A voice agent that asks about the one word it missed, instead of asking you to repeat the whole sentence. Powered by Universal-Streaming, dynamic keyterms prompting, and bounded slot validation.

THE PROBLEM • 1.4 MINUTE CONVERSATIONAL CYCLE

"Sorry, could you repeat that?" is the least informative question possible.

CONVENTIONAL VOICE AGENT

Blanket Repetition

When a voice agent mishears one token, it forces the human caller to repeat the entire 8-to-12 word sentence. Under identical acoustic noise and accent conditions, the same hard slot gets misheard again.

USER BURDEN

Average 6 to 12 words re-said per failure turn. Frustration compounds.

HUMAN CONVERSATIONAL REPAIR (SAY LESS)

Surgical Specificity

Human conversation resolves misunderstanding once every 1.4 minutes (Dingemanse et al. 2015). People always ask the *most specific* question evidence supports: naming a candidate ("Tuesday?") beats a category question ("Which day?") beats generic repetition.

THE FIX

Surgical 1-word confirmation ("Yes"). Zero wasted turns. 0.07 words re-said.

FOUR-STAGE REPAIR PIPELINE

Set membership triggers repair. Phonetics finds the candidate.

01 / BIND

Slot Binding

Universal-Streaming transcript maps onto bounded fields (day, service, surname) using AssemblyAI LLM Gateway with paired clean & garbled few-shot examples.

LLM GATEWAY

02 / CHECK

Set Membership

Confidence cannot catch a confidently-misheard token. Membership of the allowed set can. If the extracted token is outside the field's allowed set, repair triggers immediately.

SCHEMA VALIDATION

03 / MATCH

Phonetic Search

AssemblyAI gives no N-best hypotheses. Candidates are retrieved by Double Metaphone phonetic matching + fuzzy ratio against the field's known catalogue.

DOUBLE METAPHONE

04 / BIAS

Reactive Keyterms

The loop closes back to Universal-Streaming: the agent dynamically pushes narrowed candidates as `keyterms_prompt` over the live WebSocket before retry audio arrives.

STREAMING WEBSOCKET

A/B BENCHMARK HARNESS · MEASURED DECISION RECORDS

93% → 100% commit accuracy at 0.07 words re-said.

100%

Commit Accuracy (vs 93% Baseline)

Caught near-miss formatting errors (e.g. "kid's cut" vs "kids cut") silently missed by standard agents.

+7.0% DELTA

0.07

Words Re-said Per Booking (Zero Extra Turns)

Resolved with a single one-word confirmation ("Yes") instead of re-speaking full sentences.

1.00 TURNS TO RESOLVE

\$8,000

Avoided Recovery Cost Per 10,000 Inbound Calls

Based on 667 wrong bookings caught and \$7–12 contact-center human escalation recovery cost (Retell AI 2026).

AVOIDED ESCALATION

89 / 89

Unit Tests Passing & 0 Offline Cache Misses

Reproducible in 4.5s with zero network calls via committed transcript and binder evaluation caches.

REPRODUCIBLE

Conversational repair is the *missing primitive* in voice AI.

When speech recognition models meet bounded real-world workflows, intelligent repair bridges the gap between raw transcripts and flawless actions.

01 Set Membership Beats Confidence

ASR models are often confidently wrong on acoustic near-misses. Checking bounded set membership detects errors confidence misses.

02 Candidates Come From The Field

When APIs provide no alternative hypotheses, candidate offers are generated by phonetic matching against the field's allowed values.

03 Universal-Streaming Enables The Loop

Word-level metadata and mid-session dynamic keyterms updates make real-time recovery possible where black-box APIs fail.